1 Advanced techniques for improving diversification of the structures

The proposed workflow goes beyond traditional enzyme design pipelines by explicitly exploring the structural manifold of enzymatic proteins. Rather than focusing solely on reproducing known folds, the project aims to discover structurally diverse enzyme mimics that preserve catalytic functionality while introducing novel scaffold architectures. This will be achieved through a combination of generative diffusion models, sequence design networks, structural prediction models, and machine-learning driven clustering methods that enable systematic exploration of protein structure space.

Recall the proposed workflow of the AIZyme project. Step 1 is served for generating protein structures without sequences by diffusion models. This step is required for generating a promising starting pool of secondary structures. These structures are further filtered by comparing the matching score in comparison to the structure of the reference molecule (e.g., `turboPETase` or similar enzyme), and the best are kept and passed to predict sequences that correspond to these structures. This is done by employing a deep learning model `ProteinMNPP`, predicting several sequences for each structure (parameterized). Further, all these predicted sequences are passed to Step 2, involving the `AlphaFold`, for predicting their secondary structures. The bunch of produced structures is passed to Step 3, where these two set of structures, the initial one and the these generated by `AlphaFold`, are compared. Here, filtering by removing bad candidates is applied, and candidate with best parameters are selected. For this purpose, various metrics are used, such as RMSD score ranking. The obtained/survived sequences are sent to Laboratory testing (AI-zyme), and the procedure is repeated again from start.

The above designed workflow presents the backbone of our "de novo" enzyme generation, solely relying on several core components, i.e., the use of powerful tools to converge towards a pool of promising (mimic) candidate enzymes (`AlphaFold`, neural network `ProteinMPNN`, and diffusion models [1,2,3]) further processed in the laboratory hopefully confirming their catalytic properties. As the lab piece of puzzle is most time-demanding but also expensive task, one has to ensure—or at least increase the possibility—for selecting those structurally and semantically different secondary structures, sufficiently realistic in terms of catalytic characteristics to represent new enzymes with a new functionality subsequently proven in the lab testing. Shortly, to achieve this, the crucial task is to identify diverse structural scaffolds that preserve catalytic functionality but differ globally in topology.

As already mentioned, the first idea incorporates several *filtering mechanisms* to diversify the process of secondary structure-guided novel protein sequence generation. These filtering mechanisms rely on employing measuring distances in the graph space (secondary structures can be seen as 3D spatial graphs) by applying, for example, Root Mean Square (RMS) distance, or Root Mean Square Deviation (RMSD), as a metric to quantify spatial similarity between structures by measuring the average positional difference between the positions of their nucleic acids. In that way, given will be a rough estimate on how graphs really match each other in the search space of (secondary) structures. The procedure can be seen as an explicit way of filtering by involving absolute positions of structures in the search space relative to the referent enzyme (e.g., `turboPETase`). However, this usually may not be sufficient and is likely dependent of the threshold value that indicate if two structures are similar enough so that non-referent structure can be removed. Based on it, one needs to study the sensibility of `AlphaFold` w.r.t. input sequences and their score of similarity/alignment and similarity of the output structures (secondary structure of sequences). This is currently weakly explored in the literature and might be done by considering various measures of sequential similarity between multiple sequential structures. Among many, the well-established *longest common subsequence* (LCS) problem [4] seeks for object called sequences for detecting longest possible common motifs between pool of sequences, identifying structural similarities/motifs between input sequences. There are many practical variants of the LCS problem from the literature—the LCS arc-preserving problem, longest filled subsequence problem, constrained LCS problem, or gapped longest subsequence problem [5,6,7] to name a few. Their outcome is represented by more realistic patterns commonly embedded in all input sequences. In that sense, we could empirically measure the (structural) similarity of input sequences before passing them to `AlphaFold`. If the relative LCS score is high, meaning the sequences can be well-aligned, it will produce a red flag for a necessity of increasing the diversification of sequences. Another measure that could serve here is the famous *Multiple Sequence Alignment* (MSA) problem [8,9] and its score reflecting on the size of common patterns embedding. If the need for sequential diversification is detected, the current flow will be moved back to the deep learning model (`ProteinMNPP`) by sending a request for generating more diversified sequences. Therefore, the objective here is not only to seeking for candidates similar to e.g., `turboPETase`, but to explore a structurally diverse region of the enzyme manifold that preserves catalytic geometry while allowing novel folds.

Another step in the proposed workflow which could potentially deal with an increased diversification of secondary structures before going to the lab, is the subsequent step when the `AlphaFold` has generated candidate structures of the sequences, which are then filtered out by RMSD score or another one w.r.t. the reference structure. The idea is to analyze these generated structures deeply. In essence, several *measure indicators* will be incorporated for each of these structures: pocket volume, Catalytic residue geometry, stability indicators—referring to different domination-based optimization invariant in graphs, and graph topology features/descriptors such as centrality measure, radius of graphs, density. Further, fold-level descriptors are particularly relevant, such as beta-sheet topology, helix arrangement, etc. The size of minimum dominating set [10,11], the maximum (quasi-) clique [12] are known combinatorial optimization invariants that carry on important aspects in biology, protein stabilization, etc. Note that the latter two problems represent NP-hard combinatorial optimization problems, thus, for large-scale graphs computationally challenging to be examined. However, this does not produce a substantial issue, as these problems are well studied in the literature with various effective meta-heuristic approaches designed, able to reach close-to-optimal solutions within reasonable runtimes for large-scale graphs [13,14]. Here one has to deal with controlling these solvers with accordance to the allowed (time and memory) resources, therefore will be appropriately tuned by, for example, `Optuna` tool [16]. One idea is to perform the process of feature extraction on the `HPC VEGA` in a distributed fashion, speeding up the procedure significantly. Having all these (reasonably cheap) features for each generated structure is expected to be important in identifying group of similar structures, revealing/detecting those bad ones that can be replaced by improving diversity of pool of structures. Symbolically, let us assume for each structure s, a set of relevant (d) features $x_s \in \mathbb{R}^d$ is extracted. We tend to explore the idea of consensus problem based on employing different clustering techniques in Machine learning, i.e., unsupervised learning frequently used in bioinformatics—in detecting clusters of proteins with same functionality [15] in large PPI networks. Further, if by S we denote a set of all structures we deal with, one can use clustering techniques such as spectral clustering, density-based clustering or hierarchical clustering to cluster structures in S into clusters of similar structures $\mathcal{C} = (\mathcal{C}_1, \ldots \mathcal{C}_k)$ w.r.t. similar feature values; one would here seek for the existence of the highest possible number (k) of clusters with a provided good clustering (e.g., by reporting a high enough Silhouette coefficient). For each cluster $\mathcal{C}_i$, one could uniformly sample a portion (p) of structures which would be kept for promising candidates, whilst the other structures omitted and new ones regenerated by the diffusion model. These new ones would be passed through the `ProteinMPNN` and subsequently `AlphaFold`, until a predefined number of secondary structures is generated, likely more diversified than the starting one. Afterwards, it will be moving to Step 3 of the workflow.

1.1 Other more advanced ideas for enhanced structure diversification

Another point that might improve diversification of generated structures concerns directly `AlphaFold` outputs which provides much richer information that can drive diversification. `AlphaFold` *confidence signals* such as pLDDT score, PAE matrix (Predicted Aligned Error), model ensemble variance, etc. allow detecting structural uncertainty regions, which can guide diversification. Diversification will preferentially target structurally uncertain regions identified by `AlphaFold` confidence metrics.

Another idea is *Structural perturbation strategy*: Introduce direct perturbations of structures before sequence design. This could include pocket reshaping, backbone noise injection, or loop diversification, before re-running `ProteinMPNN`. This allows exploring new folds close to the catalytic scaffold.

Active Learning With Laboratory Feedback. Instead of: generate $\rightarrow$ test $\rightarrow$ repeat, use **active learning**. Example of the workflow:

- generate enzyme candidates
- cluster them
- select maximally diverse representatives
- send small batch to lab
- update predictive model using results

Then the model learns: which structures are catalytically viable. This is a closed-loop discovery system.

Enzyme Morphing: Generate **continuous structural trajectories** between: `turboPETase` and novel folds. This explores intermediate enzyme designs. Discuss the use of GED measure as well...

Technique:

- latent diffusion interpolation
- structure morphing

Then evaluate catalytic geometry. This idea represents a novel way of differentiating between newly generated folds and those with known functionality.

One Very Strong, high impact idea – **Use protein language models**:

Examples: ESM-2, ProGen, ProtT5 [17]. These provide embedding representations. Use embedding for: clustering, diversity measurement, generative search guidance. In this way, a better clustering of structures could be generated.

1.2 Visual representation of the enhanced workflow

2 Crucial and sensitive questions

– Q0. **Number of generated structures in the workflow**. Several hundreds of thousands of EMs are possible for generation within a reasonable amount of time via our workflow. This number with this order of magnitude is already revealed as promising for discovering enzymes with novel catalytic functions in the enzyme space, see J. Yang et al. (2026) [18].

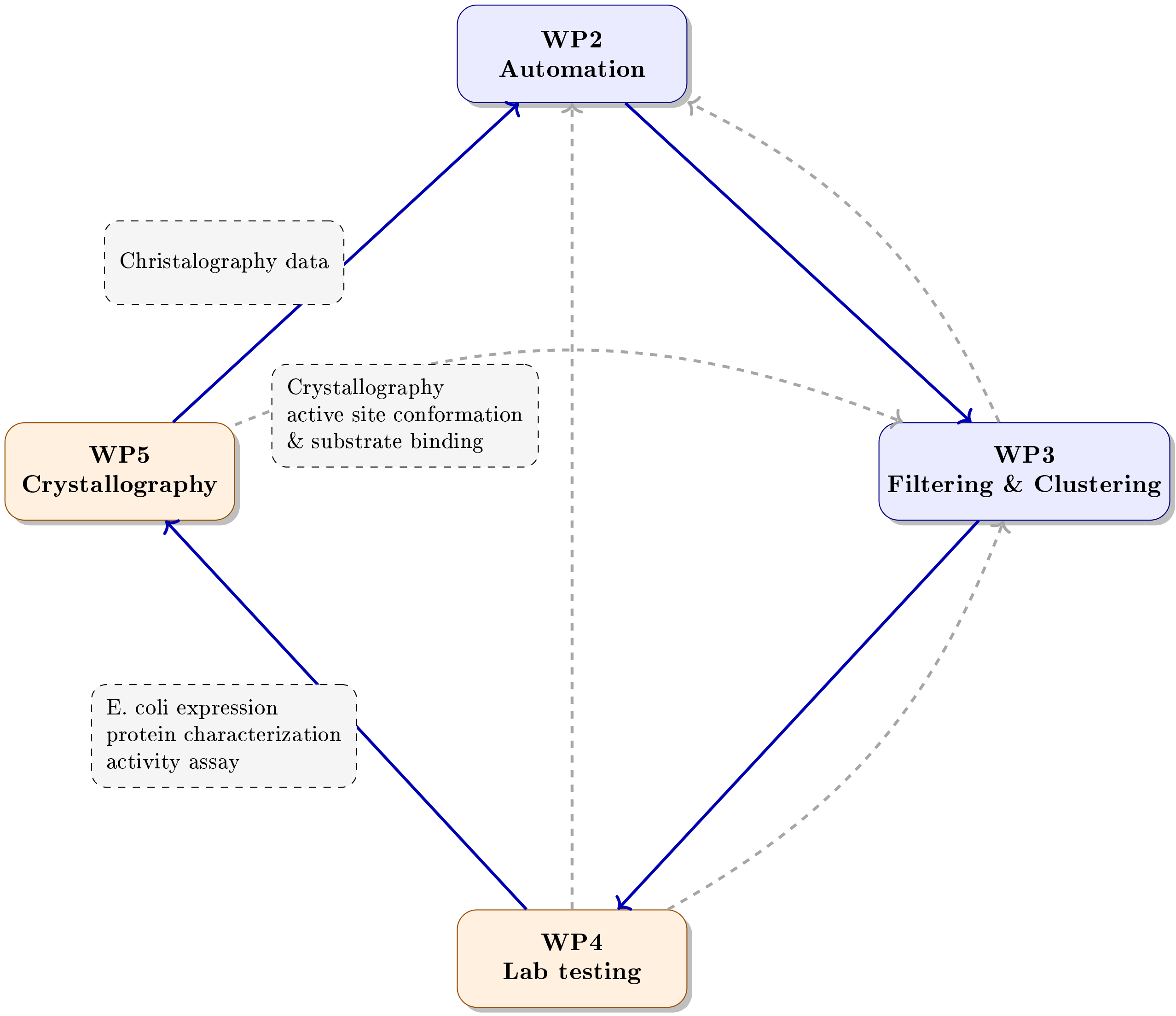
It is to be noted that many of AI-generated enzymes probably won't posses a new biocatalytic properties but, contrary, according to the generation—which is preferring the search for structures within a close distance from the selected reference enzymes—many of them will indeed represent functional enyzmes as they mimics the reference. Therefore, it remains crucial to select those enzymes which are reasonably close to the reference enzymes, but to left enough freedom to also select those further-from-reference EMs.

– Q1. **Selection of candidate enzyme mimics (EMs) for lab test**. The selection could be addressed by introducing a threshold parameters $\theta \in [0, 1]$. In essence, $100 \cdot \theta\%$ of those that are close to the reference are chosen as safe for lab test, while the rest of $(1-\theta) \cdot 100\%$ will be selected with a probability proportional to their distance from the reference enzymes (the distance here is the fitness score). According to the plan, a several thousands candidate EMs will be selected in overall for the lab testing employing this controlled random process.

– Q2. **Mitigation risks.** Several risks are detected within a project:

- None of the selected EMs posses any new catalytic property or functionality. If this occurs as the lab outcome, it may be mitigated by selecting candidate EMs in several iterations, not immediately provide a maximum number of them for analysis. In that way, we could learn from the lab outcome iteratively, by adapting the algorithmic workflow and/or the value of parameter θ several times. Another way is to include multiple reference enzymes for generating EMs. In that way, the diversity of the search towards functional enzymes might be improved yielding an enlarged possibility for discovering new EMs possessing novel catalytic properties.
- Lab risks: TODO

2.1 Xzyme workflow



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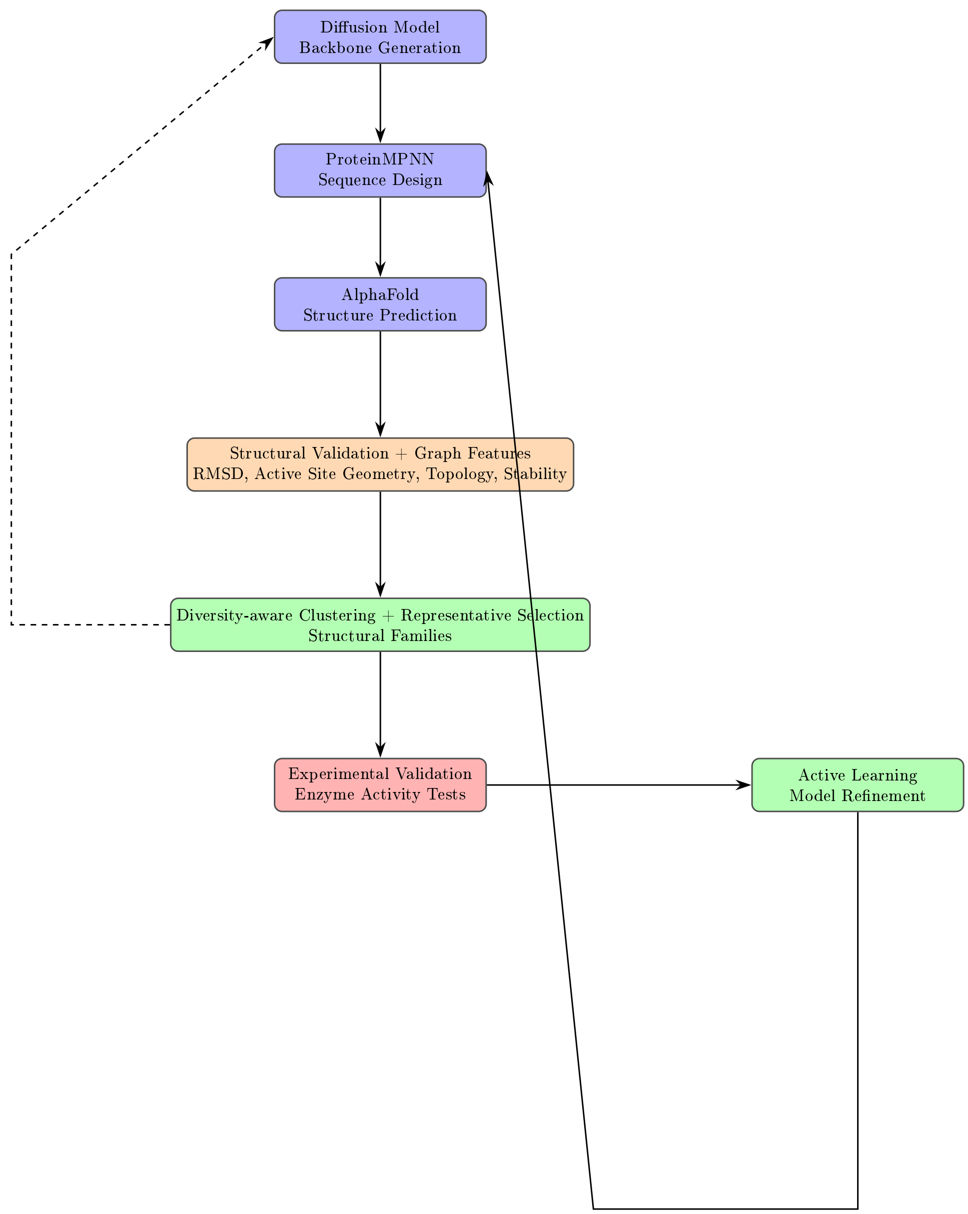

Fig. 1. Simplified AI-driven workflow for diversified de novo enzyme discovery. Core stages include generative backbone modeling, sequence design, structure prediction, structural validation with graph-based features, diversity-aware clustering, and experimental validation. Active learning feedback loops enable iterative improvement of candidate enzymes.